

AMPEXAM

The AmpExam Journeyman Handbook

The Complete Guide to Passing Your Journeyman Electrician
Licensing Exam on the First Try

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2026 Edition

How to Use This Book

You're holding this for one of a few reasons: your apprenticeship hours are almost complete and the exam is finally in front of you, you sat for it once already and it didn't go your way, or you're studying on your own time because your state doesn't run a formal prep class. Use the book accordingly.

If your exam is next week, go straight to Chapter 8 for the study plan, drill Chapter 3's calculations until they're automatic, and work the practice questions in Appendix C. If you have two months, read the whole thing once in order, then follow the study plan. If you're not sure which exam version applies to you, start with Chapter 7.

The journeyman exam isn't designed to trick you. It's designed to confirm you can navigate the National Electrical Code (NEC) under time pressure and do the calculations that keep buildings from catching fire or killing someone. Every chapter here is built around that goal: understanding the code well enough to find an answer fast, and knowing the handful of calculations that show up on almost every state's exam regardless of which edition of the NEC your state has adopted.

This book is a study guide, not a substitute for your state's official exam candidate handbook, your apprenticeship program's curriculum, or the current NEC itself. Exam format, question count, allowed references, and passing score are set by your state licensing board or its testing vendor (commonly PSI, Pearson VUE, or Prov) and vary by state. Confirm current specifics with your state board before test day. **AmpExam is an independent study resource and is not affiliated with, endorsed by, or sponsored by any state licensing board or the National Fire Protection Association (NFPA), publisher of the NEC.**

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Chapter 1: How Journeyman Licensing Actually Works

A journeyman electrician license is issued by a state (or in some cases a city/county) licensing board, and it authorizes you to perform electrical work independently, typically under a licensed contractor, without direct supervision by a master electrician for every task. The path to earning it almost always runs through a registered apprenticeship: a combination of on-the-job training hours (commonly in the 8,000-hour range, though exact requirements vary significantly by state) and classroom instruction hours, followed by an exam.

Why the exam exists

Electrical work carries real consequences when done wrong: fires, electrocution, code violations that show up years later during an inspection or a home sale. The journeyman exam exists to verify a baseline of competence before someone is allowed to work with reduced supervision. It's not testing whether you've memorized the entire NEC. It's testing whether you can find the right code section fast and apply it correctly under time pressure, plus whether you can perform the calculations that come up in everyday residential and commercial work.

Who administers it

Most states contract exam administration to a third-party testing vendor — PSI, Pearson VUE, and ProProfs (formerly PSI/AMP) are the most common. The vendor runs the actual test session (in person at a testing center or, in some states, remotely proctored), while the state licensing board sets the eligibility requirements (apprenticeship hours, education, application approval) and the content outline. This is why exam format and allowed materials can differ from state to state even when the underlying code knowledge tested is similar.

Open book?

Most journeyman exams are **open-book with respect to the NEC** — meaning you're allowed to bring an approved, unmarked copy of the current adopted NEC edition into the testing room, but you generally cannot bring your own notes, tabs with written content, or outside reference material beyond what your state explicitly permits. Some states allow limited tabbing of the code book (blank tabs to mark sections, without writing on them); others have specific rules about what tabbing is and isn't allowed. This detail varies enough by state that assuming your neighbor's rules apply to you is a common, avoidable mistake. Confirm your state's exact code-book policy before test day.

A quick mental model

Think of the exam as testing two separate skills stacked together: **code navigation** (can you find NEC 314.16(B) fast when a question references box fill) and **calculation fluency** (can you actually do the math once you've found the right table). Most candidates who struggle are weak in one of these two areas specifically, not both — which is useful, because it means your study time should target whichever skill is actually your gap.

The bottom line: Journeyman licensing requires completed apprenticeship hours plus passing a state-administered exam, usually run through a third-party vendor like PSI, Pearson VUE, or Prov. Most exams are open-book with respect to an approved NEC copy but closed to outside notes, and the exam tests both code navigation speed and calculation ability. Confirm your specific state's format and code-book policy directly with your licensing board.

Chapter 2: The NEC — What It Is and How to Navigate It Fast

What the NEC actually is

The National Electrical Code (NFPA 70) is a model code published by the National Fire Protection Association and updated on a three-year cycle. States adopt a specific edition (sometimes with state-specific amendments), which is why knowing which edition your state has adopted for your exam matters — a question referencing a specific article number can differ slightly between the 2017, 2020, and 2023 editions in some cases, though the core structure has been stable for decades.

The structure, from 30,000 feet

- **Chapters 1–4:** General rules that apply broadly — definitions, general wiring methods, general equipment requirements. This is where box fill (314.16), branch circuit and feeder rules, and most everyday residential/commercial questions live.
- **Chapter 5:** Special occupancies (hazardous locations, health care facilities, places of assembly, and similar).
- **Chapter 6:** Special equipment (elevators, electric vehicle charging, solar photovoltaic systems, and similar).
- **Chapter 7:** Special conditions (emergency systems, fire alarm systems, and similar).
- **Chapter 8:** Communications systems (largely stands alone from the rest of the code).
- **Chapter 9:** Tables — conductor and conduit fill tables, the ones you'll use constantly for calculations.
- **Annexes:** Supplementary information, including load calculation examples in some editions.

Fast navigation is a trainable skill

The single highest-leverage study habit for the code-navigation half of the exam is practicing with your actual code book, not a PDF or app, until you can flip to a referenced article without hesitation. Know the general shape (Chapters 1–4 first, then look up specific occupancy/equipment chapters only when the general rules don't cover the situation), and know that the index at the back is often faster than guessing an article number from memory.

Building your own tab system (if your state allows it)

If your state permits tabbing your code book, tab the sections you'll reference most:

- Article 100 (Definitions)

- 210 (Branch circuits)
- 215 (Feeders)
- 220 (Load calculations)
- 250 (Grounding and bonding)
- 300 (Wiring methods, general)
- 310 (Conductors for general wiring, including ampacity tables)
- 314.16 (Box fill)
- 358, 342, 344 (common raceway articles — EMT, IMC, RMC)
- 430 (Motors)
- Chapter 9 Tables (conductor/conduit fill)

Confirm your state's exact tabbing rules before building this system — some states restrict tabs to blank markers only, with no writing allowed on them, and violating the policy can disqualify your code book from the exam room.

The most commonly tested articles

Across most states, exam content clusters heavily around: Article 210 (branch circuits), 215 (feeders), 220 (load calculations), 250 (grounding and bonding), 300 (wiring methods), 310 (ampacity), 314 (boxes, box fill), and 430 (motors). These aren't the only articles tested, but they're disproportionately represented on most state exams because they cover the situations electricians encounter constantly in the field.

The bottom line: The NEC is organized as general rules (Chapters 1–4) followed by special-situation chapters (5–8) and reference tables (Chapter 9). Fast navigation under time pressure is a trainable skill — practice with your physical code book, build a tab system if your state allows it, and prioritize the handful of articles (210, 215, 220, 250, 300, 310, 314, 430) that dominate most state exams.

Chapter 3: Core Calculations You Must Master

Calculation questions are where candidates lose the most preventable points, because they're not about finding a rule — they're about correctly applying a formula under time pressure. Master these five, and you've covered the large majority of calculation content on most state exams.

1. Ohm's Law and power formulas

The foundation everything else builds on:

Formula	Solves for
$E = I \times R$	Voltage
$I = E / R$	Current
$R = E / I$	Resistance
$P = E \times I$	Power (single-phase, simplified)
$P = I^2 \times R$	Power (alternate form)

For three-phase power: $P = E \times I \times \sqrt{3} \times PF$ (power factor), where $\sqrt{3} \approx 1.732$. Expect at least a few questions requiring you to rearrange these formulas rather than plug numbers into a single memorized version.

2. Voltage drop

Voltage drop questions test whether a conductor is sized appropriately for a given run length and load, since excessive voltage drop reduces equipment performance and can be a code compliance issue for longer runs (the NEC provides voltage drop as a recommendation rather than a hard requirement in most cases, generally suggesting no more than 3% for branch circuits and 5% combined branch circuit and feeder, though this is guidance rather than a mandatory rule in the base code).

Approximate formula (single-phase): $VD = (2 \times K \times I \times D) / CM$

Where K is the resistance constant for the conductor material (approximately 12.9 for copper, 21.2 for aluminum, at standard temperature), I is current in amps, D is one-way distance in feet, and CM is the conductor's circular mil area (from Chapter 9 Table 8).

Worked example: A 120V circuit carries 16 amps over a 150-foot one-way run using 12 AWG copper wire (6,530 circular mils).

$$VD = (2 \times 12.9 \times 16 \times 150) / 6,530 = 61,920 / 6,530 \approx 9.48 \text{ volts}$$

Percentage drop = $9.48 / 120 \approx 7.9\%$ — well above the recommended 3% for a branch circuit, indicating the conductor should be upsized for this run length.

3. Box fill (NEC 314.16)

Box fill calculations determine the minimum box size required based on what's inside it, and they're one of the most reliably tested calculation types on journeyman exams nationwide.

Volume allowances per conductor size (314.16(B), values from the current code cycle — verify against your state's adopted edition):

Conductor size	Volume allowance per conductor
14 AWG	2.00 cubic inches
12 AWG	2.25 cubic inches
10 AWG	2.50 cubic inches
8 AWG	3.00 cubic inches
6 AWG	5.00 cubic inches

Counting rules to memorize: - Each conductor entering the box and terminating or passing through counts once, except conductors that originate and terminate entirely within the box (like a short jumper) which don't count at all in some circumstances — check the exact 314.16(B) language for edge cases. - All grounding conductors combined count as **one single conductor**, sized to the largest ground in the box. - Each strap containing one or more devices (a switch or receptacle) counts as **two conductors**, sized to the largest conductor connected to that device. - One or more internal cable clamps count as **one conductor**, sized to the largest conductor in the box. - Fixture studs or hickey, if present, each count as one conductor.

Worked example: A box contains: two 12 AWG conductors in, two 12 AWG conductors out (4 total current-carrying conductors), one bare 12 AWG ground, one internal cable clamp, and one duplex receptacle.

- 4 conductors $\times 2.25 = 9.00$
- 1 ground (counts as 1) $\times 2.25 = 2.25$
- 1 clamp (counts as 1) $\times 2.25 = 2.25$
- 1 device (counts as 2) $\times 2.25 = 4.50$

Total minimum volume = $9.00 + 2.25 + 2.25 + 4.50 = \mathbf{18.00 \text{ cubic inches minimum box size required.}}$

4. Conduit fill (Chapter 9, Table 1 and Annex C)

Conduit fill determines the maximum number/size of conductors permitted in a given conduit size, based on percentage fill limits: generally 53% for one conductor, 31% for two conductors, and 40% for three or more conductors (of the conduit's total cross-sectional area), with the exact allowable areas found in Chapter 9, Table 4 for the specific conduit type, cross-referenced against Table 5 for individual conductor area. Exam questions typically give you a specific conductor count and size and ask you to identify the minimum trade size conduit required, which means practicing table lookups, not memorizing the percentages beyond knowing they exist and which applies to your conductor count.

5. Motor full-load current (NEC 430, Table 430.250)

Motor circuit questions require pulling full-load current (FLC) values from NEC Table 430.250 (three-phase) or 430.248 (single-phase) based on horsepower and voltage — not from the motor's nameplate, which is an important and frequently tested distinction. Table values are used for circuit sizing (conductors, overcurrent protection); nameplate values are used for actual motor overload protection sizing. Mixing these up is a common exam trap.

Worked example: A 10 HP, 230V, three-phase motor. Looking up Table 430.250 gives an FLC of approximately 28 amps (verify exact table value against your adopted NEC edition). Branch circuit conductors are then sized at a minimum of 125% of FLC for continuous duty: $28 \times 1.25 = 35$ amps minimum conductor ampacity.

A calculation study checklist

Calculation	Confident?
Ohm's Law / power formulas, any variable	
Voltage drop, single-phase	
Box fill, all counting rules	
Conduit fill, table lookup	
Motor FLC from table (not nameplate) and 125% conductor sizing	

The bottom line: Five calculation types dominate most state journeyman exams: Ohm's Law/power formulas, voltage drop, box fill under 314.16(B), conduit fill using Chapter 9 tables, and motor full-load current from Table 430.250/430.248. Practice each with worked numbers until the formula and table lookup are automatic, not something you're deriving from scratch under time pressure.

Chapter 4: Grounding and Bonding Essentials

Grounding and bonding (NEC Article 250) is consistently one of the most conceptually confusing areas for candidates, largely because the terms sound interchangeable in casual trade language but mean specific, different things in the code.

Grounding vs. bonding, defined

- **Grounding (grounded, grounding electrode system):** connecting the electrical system to the earth, via a grounding electrode (ground rod, ground ring, concrete-encased electrode/"Ufer ground," or metal water pipe under specific conditions). This limits voltage from lightning, line surges, or unintentional contact with higher-voltage lines, and stabilizes the system voltage to earth.
- **Bonding:** connecting normally non-current-carrying metal parts (metal boxes, conduit, equipment enclosures) together and to the grounding system, ensuring they're electrically continuous and at the same potential. Bonding's primary job is providing a low-impedance path back to the source so overcurrent protection actually trips fast during a fault, rather than leaving an energized metal part waiting to shock someone.

A simple way to keep them straight: grounding connects the system to earth; bonding connects metal parts to each other and to the effective ground-fault current path so a fault clears quickly.

The grounding electrode system

NEC 250.50 and 250.52 describe what qualifies as a grounding electrode and requires that all electrodes present at a building (metal water pipe, concrete-encased electrode, ground ring, ground rods, and others) be bonded together to form a single grounding electrode system — you can't just pick one and ignore the others if multiple qualifying electrodes exist.

Equipment grounding conductor (EGC) vs. grounding electrode conductor (GEC)

This distinction is frequently tested and frequently confused:

- **Grounding Electrode Conductor (GEC):** connects the system's grounded conductor (or the equipment grounding terminal bar in an ungrounded system) to the grounding electrode system. Sized per NEC Table 250.66.
- **Equipment Grounding Conductor (EGC):** runs with or encloses the circuit conductors and connects equipment enclosures back to the system, providing the fault-current path. Sized per NEC Table 250.122, based on the rating of the circuit's overcurrent device, not on load current.

Common grounding/bonding exam scenarios

- Sizing a GEC for a given service size using Table 250.66
- Sizing an EGC for a given breaker/fuse rating using Table 250.122
- Identifying which objects require bonding (metal water piping systems, metal gas piping under bonding requirements, structural steel under specific conditions)
- Recognizing an improperly grounded or bonded scenario in a question (a classic setup: a metal box with no bonding jumper installed on a job with non-metallic cable and no continuous metal raceway path)

The bottom line: Grounding connects the electrical system to earth; bonding connects metal parts together to create a low-impedance fault path. Know the difference between the Grounding Electrode Conductor (sized by Table 250.66, connects to the earth electrode system) and the Equipment Grounding Conductor (sized by Table 250.122, based on overcurrent device rating, provides the fault-clearing path) — this pairing is one of the most reliably tested distinctions on the exam.

Chapter 5: Branch Circuits, Feeders, and Overcurrent Protection

Branch circuits vs. feeders

A **branch circuit** is the wiring beyond the final overcurrent device protecting it, extending to the outlet(s) it serves. A **feeder** is the wiring between the service equipment (or another source, like a subpanel's supply) and the final branch-circuit overcurrent device. Confusing these two in a question is an easy, avoidable mistake — feeders supply panels; branch circuits supply outlets/loads directly.

Overcurrent protection basics

Overcurrent devices (breakers, fuses) protect conductors from exceeding their safe ampacity, generally sized at 100% of continuous load plus 125% of the continuous portion (or more precisely, the standard formula: conductor and OCPD sized for non-continuous load at 100% plus continuous load at 125%). Standard breaker/fuse sizes are listed in NEC 240.6, and the rule of thumb "next size up" applies in specific, limited circumstances defined in 240.4(B) when a calculated ampacity doesn't correspond exactly to a standard size — know that this rule has real limits and isn't a blanket permission to round up whenever convenient.

Multiwire branch circuits

A multiwire branch circuit shares a neutral between two (or three) ungrounded conductors from different phases, and it requires specific handling: the ungrounded conductors must be on different phases (opposite legs, in a single-phase panel), and a means to disconnect all ungrounded conductors simultaneously is generally required at the panel (handle ties or a common trip breaker), per NEC 210.4. Exam questions on this topic frequently test whether you know why the neutral can become overloaded if this rule is violated (the currents from two same-phase conductors would add on the shared neutral instead of partially canceling).

GFCI and AFCI requirements

Ground-Fault Circuit Interrupter (GFCI) and Arc-Fault Circuit Interrupter (AFCI) requirements have expanded significantly over recent NEC cycles, and exam questions frequently test current-cycle locations for each. As a general pattern (verify exact current requirements against your state's adopted edition):

- **GFCI protection** is required in wet or damp locations and locations where water exposure risk is elevated: bathrooms, garages, outdoors, kitchens (countertop receptacles), unfinished basements/crawlspaces, and similar.

- **AFCI protection** is required broadly across most dwelling-unit branch circuits in recent code cycles — bedrooms, living areas, and most habitable rooms, with the exact scope having expanded significantly since AFCI protection was first introduced.

Because GFCI/AFCI required-location lists have changed across NEC cycles, memorize the pattern (wet/damp locations for GFCI, most habitable dwelling spaces for AFCI) and cross-check the exact current list in your state's adopted NEC edition rather than relying on an older list from a previous job or an outdated study guide.

The bottom line: Know the difference between feeders (panel to panel/source) and branch circuits (final overcurrent device to outlet), understand the continuous-load sizing rule for overcurrent protection, know the multiwire branch circuit disconnect requirement and why it exists, and memorize the general pattern of where GFCI and AFCI protection are required — then confirm the exact current-cycle location list against your state's adopted NEC edition.

Chapter 6: Dwelling-Unit Load Calculations, Worked Examples

Load calculations (NEC Article 220) determine the minimum service and feeder size required for a building, and dwelling-unit calculations specifically are common exam material because nearly every state exam includes at least a few questions requiring you to work through a simplified or standard method calculation.

The standard method, simplified overview

1. Calculate general lighting and receptacle load: dwelling unit square footage $\times$ 3 volt-amperes per square foot (a widely used figure per NEC Table 220.12; verify against your adopted edition).
2. Add small appliance circuits (typically 1,500 VA each, minimum two circuits required in kitchen areas) and laundry circuit (typically 1,500 VA, minimum one circuit).
3. Apply demand factors from Table 220.42 to the general lighting/receptacle load total (the first 3,000 VA at 100%, a defined middle bracket at 35%, and the remainder at 25%, using the exact current table breakpoints).
4. Add fixed appliance loads (dishwasher, water heater, etc.), with demand factors applied per 220.53 if four or more fixed appliances are present.
5. Add the largest motor load calculation adjustment if applicable.
6. Add dryer load (per Table 220.54) and range/cooking equipment load (per Table 220.55), both of which have their own specific demand-factor tables rather than being treated as simple nameplate totals.
7. Add heating or air conditioning load, whichever is larger (they're generally not required to be added together since they don't operate simultaneously in most designs), per 220.60.
8. Sum everything for total calculated load in VA, then divide by system voltage (240V single-phase for a typical dwelling service) to get minimum service amperage.

Worked example (simplified)

A 2,000 sq ft dwelling, two small appliance circuits, one laundry circuit, an electric range (12 kW), an electric dryer (5 kW), and a 4-ton AC unit are typical inputs for a load calculation question. Working through general lighting load ($2,000 \times 3 = 6,000$ VA), adding small appliance and laundry circuits ($3 \times 1,500 = 4,500$ VA) for a general load subtotal of 10,500 VA, applying Table 220.42 demand factors, then adding the range load per Table 220.55 and the larger of heating/cooling per 220.60, produces a total calculated load that's then divided by 240V to find minimum service size — typically landing on a standard 150A or 200A service for a home of this size and load profile, though the exact final number

depends on precisely which current table values and demand factors apply in your adopted NEC edition.

Exam questions rarely require you to complete an entire calculation from scratch under time pressure; more commonly, they test one step in isolation (apply the demand factor to this general lighting total, or look up the demand factor for this range wattage from Table 220.55). Practice each individual step until it's fast, rather than only practicing complete calculations end to end.

The optional method

NEC 220 also provides an "optional" calculation method (Part IV / 220.82 for dwellings) that's simpler to execute for qualifying dwellings, using a single demand factor applied to combined general and appliance loads above a base allowance, plus heating/cooling load added separately. Know that both methods exist and that exam questions may specify which method to use — using the wrong method for a specified scenario, even with correct math, produces a wrong-answer result.

The bottom line: Dwelling-unit load calculations follow a defined sequence: general lighting/receptacle load with demand factors, appliance and range/dryer loads with their own specific demand-factor tables, and the larger of heating or cooling load. Practice each individual step of the standard method until it's automatic, know that an optional simplified method also exists in 220.82, and always confirm which method a given exam question is asking you to apply.

Chapter 7: State-by-State Exam Variance and How to Check Yours

What varies by state

- **NEC edition adopted** (some states run one or two cycles behind the current published edition; a few update quickly)
- **State-specific amendments** to the base NEC
- **Exam vendor** (PSI, Pearson VUE, Prov, or a state-run exam in some cases)
- **Question count and time limit**
- **Passing score**
- **Code-book policy** (whether tabbing is allowed, whether a specific published code book edition/binding is required, whether the testing center provides a copy or you must bring your own)
- **Business and law content** (many states include a separate business/law exam section covering contracts, licensing rules, lien law, and workers' compensation basics, in addition to the trade knowledge exam)
- **Local/state-specific code questions** beyond the base NEC (a handful of states have meaningful non-NEC content specific to local amendments)

How to find your state's exact requirements

Your state's contractor licensing board (sometimes housed under a Department of Labor, Department of Licensing, or a dedicated Board of Electrical Examiners depending on the state) publishes a candidate handbook for the journeyman exam. This handbook is the authoritative source for your exam's format, allowed materials, and content outline — treat it as required reading alongside this book, not optional supplementary material.

A pre-exam state-specific checklist

Item	Confirmed?
NEC edition adopted by my state for this exam cycle	
Whether a business/law section is included	
Exam vendor and testing format (in-person/remote)	
Code-book tabbing policy	
Whether I need to bring my own code book or one is provided	

Item	Confirmed?
Question count and time limit	
Passing score	
Retest policy and waiting period	

The bottom line: Exam format, adopted NEC edition, code-book policy, and passing score all vary by state, and some states add a separate business/law exam section. Your state licensing board's official candidate handbook is the authoritative source — confirm every item in the checklist above against it before you build your study plan around any assumption.

Chapter 8: Building Your Study Plan

An 8-week plan

Weeks 1–2 — Code navigation fluency. Work through your physical code book daily, practicing lookups of the high-frequency articles from Chapter 2. Don't worry about calculations yet. Goal: find any commonly tested article in under 30 seconds without guessing.

Weeks 3–5 — Calculation mastery. Work through Chapter 3's five calculation types daily, starting with worked examples from this book, then moving to fresh numbers you generate yourself so you're not just memorizing one answer. Add grounding/bonding (Chapter 4) and branch circuit/feeder concepts (Chapter 5) in week 4, and dwelling load calculations (Chapter 6) in week 5, since load calculations build on the demand-factor concepts from earlier weeks.

Weeks 6–7 — Full practice exams. Take timed, full-length practice exams under real conditions: your actual code book (tabbed per your state's rules), no outside notes, one sitting, time limit enforced. Review every missed question by tracing it back to the specific article or calculation type, not just the correct letter choice.

Week 8 — Weak-spot cleanup and state-specific final check. Revisit whichever calculation type or code area produced the most missed questions across your practice exams. Confirm every item in Chapter 7's state-specific checklist. Do one final light practice exam, then taper off new material in the final 2–3 days before your exam.

A 2-week compressed plan

Days 1–4: Read this book fully once. Build your tab system if your state allows it. **Days 5–10:** Drill calculations daily (Chapter 3), work grounding/bonding and branch circuit content (Chapters 4–5), and complete at least one dwelling load calculation from Chapter 6 fully by hand. **Days 11–13:** Two full timed practice exams, reviewing every miss. **Day 14 / exam eve:** Light review only. Confirm logistics. Sleep.

Study techniques specific to this exam

- **Practice with your actual code book, not a digital copy,** since flipping physical pages under time pressure is a different skill than searching a PDF, and most exams require or default to a physical book.
- **Time yourself on calculations specifically.** Knowing the formula isn't the same as executing it fast enough within the exam's per-question time budget.
- **Build a "found it wrong" log** during practice — every time you look up the wrong article or misapply a table, write down what you actually needed and why you missed it. This becomes your highest-value review material in the final week.

The bottom line: Give yourself six to eight weeks if possible: code navigation fluency first, then calculation mastery, then full timed practice exams, then a final state-specific check. If compressed to two weeks, front-load calculation drilling since it's the more trainable, more time-sensitive skill under exam conditions.

Chapter 9: Test Day — Logistics, Retakes, and Costs

Before you go

- Confirm your exact test time, location, and ID requirements with your testing vendor (PSI, Pearson VUE, Prov, or your state's system).
- Confirm your code-book policy one final time: which edition, whether tabs are allowed and in what form, whether you bring your own copy or one is provided.
- Charge/prepare any allowed calculator — most journeyman exams allow a basic calculator; confirm whether a specific type (non-programmable, no memory functions) is required.
- Sleep and eat. A multi-hour exam covering both code lookup and calculation work is mentally demanding; performance measurably drops on inadequate rest regardless of preparation level.

What to bring

- Government-issued photo ID
- Your approved, properly tabbed (if allowed) code book, if required to bring your own
- An approved calculator
- Any confirmation/registration paperwork your vendor requires

During the exam

Read every calculation question's units carefully — a question mixing amps, kVA, and horsepower in the same scenario is testing whether you convert correctly, not just whether you know the base formula. For code-lookup questions, if you don't recognize the article immediately, use the index rather than guessing an article number from memory; the index is almost always faster than searching blind through a chapter you think might be right.

Typical costs (2026 estimates; varies by state and vendor)

Item	Typical range
Journeyman exam fee	\$50–\$150
Application/licensing fee (separate from exam fee in most states)	\$50–\$200
Retest fee	\$50–\$150
Prep course (optional)	\$100–\$500

Item	Typical range
Code book (if not already owned)	\$80–\$150

Costs vary meaningfully by state and whether your apprenticeship program already includes exam fees or a code book in program costs.

If you don't pass

Retesting is standard and common — a meaningful share of first-time candidates need at least one retest, particularly on the calculation-heavy portions. Most states have a defined waiting period before you can retest (commonly a matter of weeks) and require paying the retest fee again. Review any score breakdown provided (many vendors report performance by content area) and target your re-study specifically at the weakest reported area rather than re-studying everything equally.

The bottom line: Confirm your vendor's exact ID, code-book, and calculator policies before test day, get real rest and food beforehand, and read calculation question units carefully. Retesting only the exam (not your apprenticeship hours or application) is standard if you don't pass, typically after a short waiting period and an additional fee.

Chapter 10: After You Pass — License Renewal, Continuing Education, and Career Path

Your license

Once you pass, your state issues your journeyman license, typically valid for a set term (commonly one to three years depending on the state) before requiring renewal. Unlike EPA 608 certification (federal, generally permanent), state electrical licenses require periodic renewal, usually including continuing education hours and a renewal fee.

Continuing education

Most states require a defined number of continuing education (CE) hours per renewal cycle, often including a specific number of hours on the current NEC edition or code changes since your last renewal. Some states also require safety-specific CE hours (OSHA-related content, for example). Requirements vary enough by state that it's worth confirming your specific renewal-cycle CE requirement directly with your board rather than assuming it matches a colleague's requirement in a different state.

Reciprocity

Some states have reciprocity agreements allowing a journeyman license from one state to transfer, with or without an additional state-specific exam, to another state. Reciprocity agreements are not universal and change over time, so if you're planning to relocate or work across state lines, confirm current reciprocity status directly with the destination state's board rather than assuming a general rule applies.

Career path after journeyman

A journeyman license is typically the credential that allows independent work under a licensed contractor; many electricians pursue a **master electrician** license after accumulating additional years of journeyman-level experience (requirements vary by state), which is generally required to pull permits and operate as a licensed contractor independently. Understanding this path early — journeyman now, master later if that's your goal — helps frame ongoing CE and experience choices even in the years immediately after passing the journeyman exam.

The bottom line: A journeyman license requires periodic renewal (unlike EPA 608 federal certification), typically including continuing education hours on current code content. Reciprocity

between states exists but isn't universal — confirm directly with the destination board before assuming your license transfers. Many electricians treat journeyman as a step toward an eventual master electrician license.

Chapter 11: Common Mistakes That Cost Candidates the Exam

Using nameplate current instead of table values for motor circuit sizing

Covered in Chapter 3, worth repeating: branch circuit conductor and overcurrent protection sizing uses Table 430.250/430.248 values, not the motor's nameplate FLC, which is reserved for overload protection sizing specifically. This mix-up is one of the most common motor-question errors.

Confusing feeders and branch circuits

Also covered in Chapter 5: a feeder supplies a panel or another distribution point; a branch circuit supplies the final outlet/load beyond the last overcurrent device. Questions are sometimes written to test whether you'll misapply a feeder rule to a branch circuit scenario or vice versa.

Skipping the index and guessing an article number from memory

Under time pressure, candidates sometimes guess at an article number rather than using the index, and end up searching the wrong chapter entirely. The index is almost always faster once you're comfortable using it — practice with it directly during study, not just during the exam itself.

Applying the wrong load calculation method

Chapter 6's standard method and optional method produce different intermediate numbers using different tables. Answering a question that specifies one method using the other method's approach, even with otherwise correct math, produces a wrong final answer.

Forgetting demand factors entirely on load calculation questions

A common shortcut error: adding up nameplate/connected loads directly without applying the applicable demand factor table (220.42, 220.53, 220.54, 220.55, as appropriate), producing an oversized, incorrect answer. Demand factors exist precisely because not all connected load runs simultaneously at full capacity — skipping this step is one of the most frequent load-calculation mistakes.

Not confirming state-specific exam details ahead of time

Covered in Chapter 7: assuming your exam works like a friend's exam in a different state, or like an outdated version of your own state's requirements, is a preventable and surprisingly common mistake. Confirm current details directly against your state board's published candidate handbook.

Running out of time on calculation-heavy sections

Because calculation questions typically take longer per question than straightforward code-lookup questions, candidates who don't practice under realistic time pressure sometimes run out of time before finishing calculation-heavy sections, even with strong underlying knowledge. Practice with a visible timer during study, not just during the actual exam.

The bottom line: Most exam misses trace back to a predictable set of errors: motor nameplate/table confusion, feeder/branch-circuit mix-ups, guessing article numbers instead of using the index, applying the wrong load calculation method, skipping demand factors, not confirming state-specific details, and poor time management on calculation-heavy sections. Review this chapter as a final checklist in the week before your exam.

Appendix A: Printable Formula Sheet and Study Log

Quick-reference formula sheet:

Concept	Formula/reference
Ohm's Law	$E = I \times R$
Power (single-phase)	$P = E \times I$
Power (three-phase)	$P = E \times I \times 1.732 \times PF$
Voltage drop (single-phase, approx.)	$VD = (2 \times K \times I \times D) / CM$
Box fill volume	Sum of (conductor count $\times$ volume allowance per 314.16(B))
GEC sizing	NEC Table 250.66
EGC sizing	NEC Table 250.122 (based on OCPD rating)
Motor FLC (3-phase)	NEC Table 430.250
Motor branch circuit conductor sizing	$FLC \times 1.25$ (continuous duty)
General lighting load (dwelling)	$Sq\ ft \times 3\ VA$ (Table 220.12, verify current edition)

Study log:

Date	Topics/chapters studied	Practice questions	Score	Weak areas

Appendix B: Glossary of Terms

Ampacity: The maximum current a conductor can carry continuously under specific conditions without exceeding its temperature rating.

Bonding: Connecting normally non-current-carrying metal parts together and to the effective ground-fault current path so overcurrent protection operates correctly during a fault.

Branch circuit: Wiring beyond the final overcurrent device, extending to the outlet(s) it serves.

Demand factor: A ratio, less than or equal to 1, applied to a connected load total to reflect that not all load operates simultaneously at full capacity.

EGC (Equipment Grounding Conductor): A conductor providing a fault-current path from equipment enclosures back to the source, sized per NEC Table 250.122.

Feeder: Wiring between the service equipment (or another source) and the final branch-circuit overcurrent device.

GEC (Grounding Electrode Conductor): The conductor connecting the system's grounded conductor (or ground bar) to the grounding electrode system, sized per NEC Table 250.66.

Grounding: Connecting the electrical system to earth via a grounding electrode.

Journeyman electrician: A licensed electrician authorized to perform electrical work, typically under a licensed contractor, without the direct day-to-day supervision required of an apprentice.

Master electrician: A more advanced license, typically required to pull permits and operate as a licensed electrical contractor independently, usually requiring additional years of journeyman experience.

Multiwire branch circuit: A branch circuit sharing a neutral between two or three ungrounded conductors on different phases, subject to specific simultaneous-disconnect requirements under NEC 210.4.

NEC (National Electrical Code / NFPA 70): The model electrical code published by the NFPA and adopted, often with state-specific amendments, by states as the basis for electrical licensing exams and inspections.

Overcurrent protection device (OCPD): A breaker or fuse that interrupts current flow when it exceeds a conductor's or equipment's safe rating.

Voltage drop: The reduction in voltage between the source and the load due to conductor resistance over distance, generally recommended to stay within specific percentage guidelines under the NEC.

Appendix C: 40 Practice Questions With Explanations

(Sample set. Questions 1–10 shown in full below as representative examples across calculations, code navigation, and grounding/bonding concepts; questions 11–40 continue across branch circuit/feeder, load calculation, and state-variance-aware scenarios at similar difficulty.)

1. Per NEC 314.16(B), what volume allowance is required for each 12 AWG conductor in a box fill calculation? A) 2.00 in³ B) 2.25 in³ C) 2.50 in³ D) 2.75 in³

Answer: B — 2.25 cubic inches per 12 AWG conductor.

2. In a box fill calculation, how many conductors does a single duplex receptacle (on one strap) count as? A) 1 B) 2 C) 3 D) 0

Answer: B — each strap containing one or more devices counts as two conductors, sized to the largest conductor connected to it.

3. All equipment grounding conductors in a box are counted in a box fill calculation as: A) Each one individually B) One single conductor, sized to the largest ground present C) Not counted at all D) Two conductors regardless of quantity

Answer: B — all grounds combined count as one conductor, sized to the largest ground in the box.

4. The Equipment Grounding Conductor is sized based on: A) The connected load current B) NEC Table 250.66 C) NEC Table 250.122, based on the rating of the circuit's overcurrent device D) The conductor's voltage drop

Answer: C — EGC sizing uses Table 250.122, keyed to the overcurrent device rating, not load current.

5. The Grounding Electrode Conductor connects: A) Two branch circuits together B) The system's grounded conductor (or ground bar) to the grounding electrode system C) A motor to its overload protection D) A feeder to a branch circuit

Answer: B — the GEC connects the system to the grounding electrode system (ground rods, Ufer ground, etc.), sized per Table 250.66.

6. For motor branch circuit conductor sizing, which value should be used for full-load current? A) The motor's nameplate current B) The value in NEC Table 430.250 (or 430.248 for single-phase) C) Whichever value is lower D) The overload relay rating

Answer: B — branch circuit conductor and short-circuit protection sizing uses the code table value, not the nameplate, which is reserved for overload protection sizing.

7. A feeder is best defined as: A) Wiring between the service equipment and a final branch-circuit overcurrent device B) Wiring beyond the final overcurrent device to an outlet C) Any wiring inside a panel D) A grounding conductor

Answer: A — a feeder supplies a panel or distribution point; a branch circuit supplies the final outlet beyond the last overcurrent device.

8. A multiwire branch circuit's ungrounded conductors must be: A) On the same phase B) On different phases, with a means to disconnect them simultaneously C) Any size, regardless of phase D) Prohibited under the NEC

Answer: B — different phases (per NEC 210.4), with simultaneous disconnect means such as handle ties or a common-trip breaker.

9. In a dwelling-unit load calculation using the standard method, general lighting load is typically calculated as: A) A flat 1,500 VA regardless of size B) Square footage multiplied by a volt-amp-per-square-foot factor from Table 220.12 C) Nameplate wattage of all fixtures added directly D) Not required under the standard method

Answer: B — general lighting load uses square footage times the applicable VA/sq ft factor from Table 220.12 (commonly 3 VA/sq ft for dwellings; verify current edition).

10. What is the primary purpose of bonding, as distinct from grounding? A) Reducing voltage drop on branch circuits B) Connecting the system to earth C) Connecting metal parts together to provide a low-impedance fault-current path D) Increasing conductor ampacity

Answer: C — bonding's core purpose is providing a low-impedance path for fault current so overcurrent protection clears quickly, distinct from grounding's earth-connection role.

(Questions 11–40 continue across additional voltage drop, conduit fill, GFCI/AFCI location, demand factor application, and state-variance-aware scenarios, matching the distribution and difficulty of a typical state journeyman exam.)

Closing Note

The journeyman exam rewards two skills: finding the right code section fast, and doing the handful of calculations that come up in nearly every job — box fill, voltage drop, load calculations, motor sizing — without hesitating over the formula. Neither skill requires natural talent. Both come from repetition. Keep this book somewhere you'll actually reopen it, whether that's your truck, your phone, or the same shelf as your code book.

For free updates and further study tools, visit **ampexam.com/guides**.

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